Cofinal compact-window Weil positivity

A self-contained finite-matrix problem

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Abstract

We isolate one concrete lemma: for infinitely many indices in an explicit sequence, a finite bordered dual resolvent is at least one half of the identity. The matrix is built from the von Mangoldt comb, the archimedean Weil kernel, one fixed smooth comparison comb, and the reserve 5/7. The measured race is 0.606566–0.757783 for $k = 5, \dots, 12$, but no infinite-subsequence proof is known. After two internal identifications, the lemma has a classical reading as positivity of the Weil quadratic form on a cofinal sequence of compact support windows. Three further bridges—density, standard explicit-formula identification, and the zeta/RH interface—are separate open problems. This document asks for the finite-matrix lemma; it does not claim the Riemann hypothesis.

Grades. [E] means an exact finite identity or a proved elementary lemma; [C] means a sealed finite census; [O] means open. No census is extrapolated and no open statement is promoted.

1 The named sequence

For $k \geq 1$ set

$$a_k = 2^k, \qquad r_k = \lfloor \sqrt{k} \rfloor, \qquad m_k = k2^{r_k} - 1, \qquad M_k = m_k + 1, \qquad L_k^{\text{grid}} = 2M_k - 2, \tag{1}$$

and

$$\Delta_k = \frac{\log a_k}{M_k} = 2^{-r_k} \log 2, \qquad N_k = \frac{M_k}{2} = k2^{r_k - 1}. \tag{2}$$

Here M_k is even for the range of interest. The folded grid has $m_k = L_k^{\text{grid}}/2$ points

$$\theta_{k,j} = \frac{2\pi j}{L_k^{\text{grid}}}, \qquad x_{k,j} = \cos \theta_{k,j}, \qquad 1 \leq j \leq m_k.$$

The last point is its own mirror and receives half the generic fold weight.

2 The faithful fold

Write

$$T_\Delta(t) = \max\{0, 1 - |t|/\Delta\}$$

and let $\Lambda(n) = \log p$ if $n = p^\ell$ is a prime power, and 0 otherwise. The prime lags are the finite sums

$$c_{k,i}^P = -\sum_{2 \leq n \leq a_k^2} \frac{\Lambda(n)}{\sqrt{n}} \left[T_{\Delta_k}(i\Delta_k - \log n) + T_{\Delta_k}(i\Delta_k + \log n) \right], \quad 0 \leq i < M_k. \tag{3}$$

The reflected term is nonzero only when $\log n < \Delta_k$.

For completeness, define the archimedean lag kernel without a numerical quadrature convention. Put γ_E for Euler's constant. For $s \geq \Delta > 0$ let

$$A_\Delta(s) = -\int_{s-\Delta}^{s+\Delta} T_\Delta(s-w)\frac{e^{-w/2}}{1-e^{-2w}}\,dw. \tag{4}$$

For $0 \leq s < \Delta$, put $t = T_\Delta(s)$, $S_s(w) = \frac{1}{2}(T_\Delta(s-w) + T_\Delta(s+w))$, and define

$$\begin{aligned} A_\Delta(s) = & -(\gamma_E + \log\pi)t + 2\int_0^{s+\Delta} \frac{te^{-2w} - S_s(w)e^{-w/2}}{1-e^{-2w}}\,dw \\ & - t\log(1-e^{-2(s+\Delta)}). \end{aligned} \tag{5}$$

The apparent singularity at $w = 0$ is removable. Set $c^A_{k,i} = A_{\Delta_k}(i\Delta_k)$ and $c_{k,i} = c^A_{k,i} + c^P_{k,i}$.

The circulant spectral density and signed folded weights are

$$d_{k,j} = c_{k,0} + 2\sum_{i=1}^{M_k-2} c_{k,i}\cos(i\theta_{k,j}) + c_{k,M_k-1}\cos((M_k-1)\theta_{k,j}), \tag{6}$$

$$w_{k,j} = \frac{2}{L_k^{\mathrm{grid}}}(1-\cos\theta_{k,j})d_{k,j} \times \begin{cases} 1, & j < m_k, \\ \frac{1}{2}, & j = m_k. \end{cases} \tag{7}$$

Define positive measures on $[-1,1]$ by the sign split

$$\mu_k = \sum_j \max(w_{k,j},0)\,\delta_{x_{k,j}}, \qquad \nu_k = \sum_j \max(-w_{k,j},0)\,\delta_{x_{k,j}}. \tag{8}$$

Equations (1)–(8) are the faithful production order: tents, total lags, circulant density, fold, sign split. No atom-table cutoff occurs; all prime powers through a_k^2 are included.

3 The border, budget, and $R^\dagger$

The border is fixed by the following smooth comparison comb. Let $\eta = 10^{-2}$,

$$u_\ell = \ell\eta, \qquad m_\ell = 2e^{u_\ell/2}\eta, \qquad 0 < u_\ell < 2\log a_k.$$

Replace the prime-power positions and masses in (3) by (u_ℓ, m_ℓ), equivalently use

$$c^\sigma_{k,i} = -\frac{1}{2}\sum_\ell m_\ell\left[T_{\Delta_k}(i\Delta_k - u_\ell) + T_{\Delta_k}(i\Delta_k + u_\ell)\right],$$

add the same $c^A_{k,i}$, and apply (6)–(7). Retaining the signs gives a finite signed measure σ_k.

Assume the moment Gram of μ_k is positive through degree $N_k - 1$, and let $p_{k,0}, \ldots, p_{k,N_k-1}$ be its real orthonormal polynomials. Define

$$b_{k,i} = \int p_{k,i}(x)\,d\sigma_k(x), \qquad B_k^{\mathrm{bud}} = \sum_{i=0}^{N_k-2} b_{k,i}^2 + \frac{5}{7}, \qquad \beta_k = \frac{b_k}{\sqrt{B_k^{\mathrm{bud}}}}. \tag{9}$$

Thus both numerical conventions in the border are explicit: $\eta = 0.01$ and the reserve is exactly 5/7.

Here is an intrinsic definition of the dual block. Let $X_k = \mathrm{supp}(\mu_k + \nu_k)$, $P_{X_k}(x) = \prod_{\xi \in X_k}(x - \xi)$, and, for $\xi \in X_k$, let

$$u^\vee_{k,\xi} = \frac{1}{(\mu_k+\nu_k)(\{\xi\})\,P'_{X_k}(\xi)^2}.$$

Let $\Pi^{\vee}_{k,N_k-1}$ be the dressed Christoffel–Darboux projection of rank $N_k - 1$ for this reciprocal measure, and let R_k be its restriction to $Y_k = \operatorname{supp}\nu_k$. For $y \in Y_k$, set

$$(v_k)_y = \sqrt{\nu_k(\{y\})}\sum_{i=0}^{N_k-1}(\beta_k)_i p_{k,i}(y), \qquad (D_k)_{yy} = \operatorname{sgn} P'_{X_k}(y), \quad \widetilde{v}_k = D_k v_k,$$

and $\gamma_k = \|\beta_k\|^2$. The bordered dual resolvent is

$$\boxed{R^\dagger(W_k) = \begin{pmatrix} R_k^{-1} & \widetilde{v}_k \\ \widetilde{v}_k^T & 1+\gamma_k \end{pmatrix}^{-1}.} \tag{10}$$

The definition applies whenever the displayed Gram and inverse conditions hold. Their failure at infinitely many indices would itself refute the requested statement.

4 The problem in three coordinates

Problem 1 (Finite-matrix form)**.** [O] *Prove*

$$\boxed{R^\dagger(W_k) \succeq \frac{1}{2}I \quad \textit{for infinitely many } k \textit{ in the sequence } (1).}$$

This semidefinite statement, rather than a strict asymptotic margin, is the requested lemma. It has two useful equivalent coordinates.

Classical coordinate. For a compactly supported $h \in L^2(\mathbb{R})$ put $g = h * \widetilde{h}$ and $Q_{\mathrm{W}}(h) = W(g)$, the additive-log Weil quadratic form. Define

$$\lambda_*(L) = \inf_{\substack{0\neq h\in L^2(\mathbb{R})\\ \operatorname{supp} h\subset[-L,L]}} \frac{Q_{\mathrm{W}}(h)}{\|h\|_2^2}.$$

The selected support scales are

$$U_k = N_k\Delta_k = \frac{k\log 2}{2}, \qquad L_k = \frac{U_k}{2} = \frac{k\log 2}{4}. \tag{11}$$

After the internal fold/read and archimedean identifications, Problem 1 reads

$$\lambda_*(L_k) \geq 0 \quad \text{for an unbounded subsequence.}$$

Because the support spaces are nested, $\lambda_*(L)$ is nonincreasing; hence this is cofinal compact-window Weil positivity.

Spectral coordinate. Let

$$(\mathcal{B}_k)_{yi} = \sqrt{\nu_k(\{y\})}\,p_{k,i}(y), \qquad C_k = \mathcal{B}_k^T\mathcal{B}_k, \qquad u_k = \beta_k.$$

If $C_k e_{k,j} = \lambda_{k,j}e_{k,j}$ and $C_k \prec I$, then

$$q_k^\dagger := u_k^T(I - C_k)^{-1}u_k = \sum_j \frac{|\langle u_k, e_{k,j}\rangle|^2}{1-\lambda_{k,j}}. \tag{12}$$

Finite duality and Schur complementation give

$$R^\dagger(W_k) \succeq \tfrac{1}{2}I \quad\Longleftrightarrow\quad \begin{pmatrix} I - C_k & u_k \\ u_k^T & 1 \end{pmatrix} \succeq 0 \quad\Longleftrightarrow\quad q_k^\dagger \leq 1 \tag{13}$$

under the stated nondegeneracy assumptions. Thus the sharp internal task is to bound the spectral measure of the explicit border vector near the saturating edge $\lambda = 1$.

5 What is known

5.1 Unconditional compact-support zones

For an autocorrelation $g = h * \widetilde{h}$ one has $\widehat{g} = |\widehat{h}|^2 \geq 0$ [E]. Bombieri's classical small-support theorem gives strict Weil positivity when the length of the support interval of h is $U < \log 2$. A recent computational preprint reports the certified extension

$$Q_{\mathrm{W}}(h) \geq 8.9 \cdot 10^{-18} \|h\|_2^2 \quad \text{for} \quad \operatorname{supp} h \subset [-0.8, 0.8],$$

or autocorrelation support $U \leq 1.6$ [C]. This recent bound is not being treated as a classical theorem.

By (11), $U_5 = 1.732868 > 1.6$. Every measured selected window therefore lies beyond both unconditional zones. The crossing is part of the point: the finite race below is not explained by prime blindness or by the $U \leq 1.6$ certificate.

5.2 Finite selected-window census

k	m_k	N_k	Δ_k	U_k	race $q_k^\dagger$	raw norm bound	top overlap
5	19	10	.1732868	1.732868	.675060	4.825	$9.53\,10^{-3}$
6	23	12	.1732868	2.079442	.694907	14.227	$2.22\,10^{-6}$
7	27	14	.1732868	2.426015	.628736	47.207	$8.89\,10^{-4}$
8	31	16	.1732868	2.772589	.606566	93.728	$5.05\,10^{-4}$
9	71	36	.0866434	3.119162	.635369	724.586	$1.94\,10^{-5}$
10	79	40	.0866434	3.465736	.757783	1593.258	$4.15\,10^{-6}$
11	87	44	.0866434	3.812309	.715782	2755.648	$5.10\,10^{-6}$
12	95	48	.0866434	4.158883	.693564	4991.649	$1.54\,10^{-6}$

All eight races lie in $[0.606566, 0.757783]$ and are below 1 [C]. The spectral identity (12) agrees with the direct solve to $3.85 \cdot 10^{-14}$ and with the independently sealed race object to $1.89 \cdot 10^{-14}$ [C]. None of these numbers proves an infinite subsequence.

5.3 Exact identities available to an attack

Identity	One-line proof
$\Delta_k = 2^{-r_k} \log 2$ and $U_k = k \log 2/2$ [E]	Insert $a_k = 2^k$ and $M_k = k2^{r_k}$ into $\Delta_k = \log(a_k)/M_k$ and $U_k = N_k \Delta_k$.
$\#\{k : \lfloor\sqrt{k}\rfloor = r\} = 2r + 1$ [E]	The block is the integer interval $[r^2, (r+1)^2)$.
$\widehat{h * \widetilde{h}} = \lvert\widehat{h}\rvert^2$ [E]	Use the convolution theorem and $\widehat{\widetilde{h}} = \overline{\widehat{h}}$.
Spectral race identity (12) [E]	Diagonalize the real symmetric matrix C_k.
Schur equivalence in (13) [E]	Take the Schur complement of $I - C_k$; finite particle–hole duality supplies the first equivalence.
$\lambda_*(L)$ is nonincreasing [E]	The admissible support subspaces are nested as L increases.
Two-band estimate [E]	For $P_\theta = \mathbf{1}_{(\theta,1)}(C_k)$, split (12) and bound the two denominators by $1 - \theta$ and $1 - \lambda_{\max}(C_k)$.

6 What does not work

1. **A finite atom table.** The apparent wall at anchors 631/641 was exactly the cutoff $4 \cdot 10^5$ lying between their squares. The mathematical source requires every prime power through a_k^2.

2. **A fixed prefix.** Below the first prime-tent index $J_P = 2^{r_k} - 1$, MAIN and a world with the prime comb deleted agree to machine precision. Such a prefix is useful anatomy but is prime-blind and cannot be the mincut.

3. **Raw operator-norm or Gershgorin bounds.** The norm majorant grows from 4.825 to 4991.649 while the actual race stays below 0.758. It discards the small overlap with the dangerous eigenspaces. Gershgorin lower bounds are negative on all eight measured rows.

4. **The best two-band bound without source control.** Optimizing over every spectral cut certifies $k = 5$ with $0.910 < 1$, but exceeds 1 for $k = 6, \ldots, 12$. The projector identity is a reduction, not an estimate.

5. **One fixed-r plateau.** The spacing Δ_k is constant on a block, but each block has only $2r + 1$ indices. No argument confined to one block can prove "infinitely many k".

6. **A Weyl comparison with mismatched depths.** The archimedean baseline used in the prefix race and the full matrix live on different polynomial spaces. Extending the former to full depth already contains the desired positivity problem.

7. **Zero-density, Li, or Bombieri–Lagarias as a shortcut.** These are valuable alternate coordinates and diagnostics. They do not supply positivity for every compactly supported autocorrelation; proving all required inequalities is again the hard criterion.

7 Numerical provenance

Every non-definitional decimal and campaign constant above is tied to a sealed source. Commit prefixes identify the immutable record used here.

Numbers	Sealed source
$\eta = 0.01$; reserve 5/7	r243 source, corpus commit `36211146`; budget identity resealed r362, `62ca2bfa`
$4 \cdot 10^5$, anchors 631/641, complete-comb repair	r458, `b315ead4`
$U < \log 2$; $U \leq 1.6$; $8.9 \cdot 10^{-18}$; selected U_k, L_k values	r461 literature/geometry audit, `6f5ea563`
race values for $k = 5, \ldots, 12$ and $[0.606566, 0.757783]$	r459 full-comb cleanup, `43ea44ba`
raw bounds, overlaps, 0.910, $3.85 \cdot 10^{-14}$, $1.89 \cdot 10^{-14}$	r460 spectral race proof, `f915dd5e`
(S, N) pins, $2 \cdot 10^{-12}$, census eight	r463 faithful-fold repair, `8f81005f`

8 Verification anchors and exact claim boundary

The finite algebra has independent formal anchors:

- `masterCap_posSemidef_iff_Rdagger_ge_half` and `selectedWindowConeSemidef_implies_A_cap_posSemidef`;

- the compound premise `FrequentlySelectedInternalMincut`, which visibly contains `SelectedACapPsdImpliesPlainReads`;

- the proved conditional endpoint `internal_weil_nonneg_of_frequently_selected`;
- sealed checkers for the full-comb race, spectral reduction, narrow-band translation, end-to-end gap map, and faithful fold.

The faithful-fold pins at $k = 5, 9, 10$ have respectively $(S, N) = (19, 10), (71, 36), (79, 40)$ and reproduce the first three positive-measure Hankel entries within $2 \cdot 10^{-12}$ [C]. The formal census after exposing every known obligation is eight uses of `sorry`; this is documentation of gaps, not evidence.

Internal open steps. The implication from the selected cap matrix to the native three-channel read remains the named proposition `SelectedACapPsdImpliesPlainReads` [O]. The exact archimedean lag/read identification also remains open in the formal path [O].

External bridge 1: density and continuity. Extend positivity from native dyadic piecewise-linear autocorrelations to the complete admissible Weil test class [O]. This is a genuine analytic density/continuity theorem.

External bridge 2: explicit-formula identification. Identify the continued three-channel functional, with matching normalizations, with the standard Guinand–Weil explicit formula, including its spectral/zero side [O].

External bridge 3: zeta interface. Formalize the standard Weil criterion against the actual Riemann zeta function and its Riemann-hypothesis predicate [O].

Boundary. Problem 1 is the lemma being offered. It is open. Even a proof of that lemma must still pass the two internal identifications and the three separately listed external bridges before it can participate in an RH proof. The present document proves no statement about the zeros of ζ.

References

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